



CCNP: Configuring EBGP On Multiple Autonomous Systems And Protocols

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Purpose

BGP is a vital protocol for understanding how the internet works, especially how it sends information from various independent services. Its importance cannot be overstated, and issues that are found with various online services that seem huge in their severity, can to some degree be attributed to a poorly setup BGP protocol. The rather loose education this lab provides is thus hugely important for employment in the networking industry, as well as useful for deepening students understanding of internet protocols.

Background

BGP is a highly important protocol for large networks and especially ISP's who are practically required to use it. Famed for being whipped up in an afternoon by Yakov Rekhter, Len Bosack, and Kirk Lougheed during an IETF meeting in Austin, Texas, on two napkins gaining it the epithet "the two-napkin routing protocol." Within the year, BGP version 1 is released with internet memo RFC 1105, and was soon updated in 1990, the year after its first release, and again the following years in 1991 and 1994, leading to the modern version of BGP widely used today BGP version 4. Today it is the agreed upon inter-domain routing protocol for the internet as a whole and has proven itself to be both durable and flexible in its workings.

BGP has two main systems, eBGP and iBGP, or External Border Gateway Protocol, and Internal Border Gateway Protocol. eBGP allows communication between two different networks, forming adjacencies between two different Autonomous Systems, whereas iBGP allows for communication within said Autonomous System. This is of course a drastic simplification, but it is good enough for this lab.

Other important parts of understanding for this lab are the various gateway protocols that we'll be using. Firstly, there's OSPF, or Open Shortest Pathway First, which employs link-state routing, allowing it to build a routing table and adjust to quick changes in the topology, such as various ports going up or down. As a result of this however, the scaling of this type of protocol can often be costly in terms of power and bandwidth, the constant updates required using up resources with a hunger not to be dismissed. There are various ways to mitigate this however, through methods like Multi-Area OSPF, which splits up the network into smaller independent groups that only advertise their constant status changes to others within the group, before sending out a general message to the other groups. This helps preserve resources, and scale the program, although do note that this is still

an Internal Gateway Protocol (IGP), which means that it is not of use when going to other protocols.

EIGRP, or Enhanced Interior Gateway Routing Protocol is a Cisco proprietary protocol available for use only on Cisco routers. Its main draw is the easing of an administrator's workload, in one very important way. In normal networks, each router holds the routing information for themselves in the network, however, if a packet ever comes to the router and said router doesn't have a valid path to the destination, the router simply drops the packet. This means that individual routers all need to be configured much more carefully, leading to much more tedium. EIGRP bypasses this by having separate routers share their paths between themselves. The protocol then builds both a Neighbor, and Topology table, one for physical connections, the other for routes learned from other routers respectively. Interestingly enough, the Topology table, often sent to the routers Routing table for use, when requesting information, only receives the changes in topology since the last request, and not the whole routing table once again.

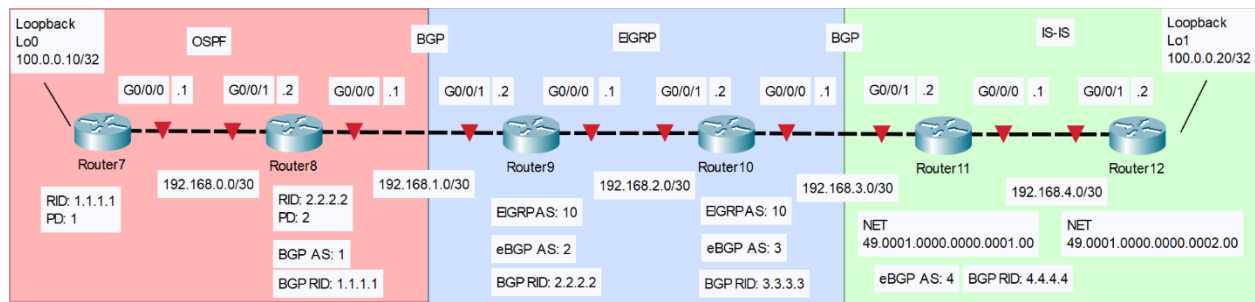
IS-IS, also known as Intermediate System to Intermediate System, uses link-state routing protocol, flooding the network with information which each IS-IS router uses to build its own link-state database (LSDB). Using the database, packets are then forwarded through the path calculated to be most efficient to their destination. Beyond this IS-IS also uses what are known as network entity titles (NET), of which the simplest format is still 10 octets long. Although it can be rather unwieldy to setup, its use as a network backbone persists to the modern day.

Procedure

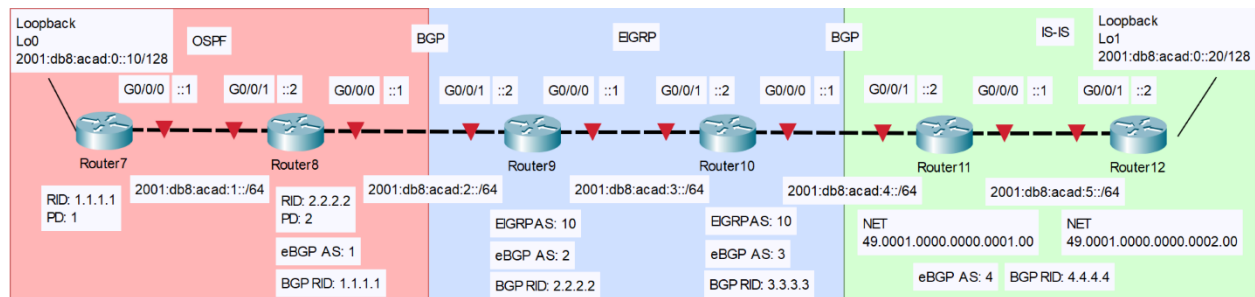
Topology

The topology consists of three autonomous systems running on six separate routers, two routers per system, plus two loopbacks on the ends of the networks. The protocols running are OSPF on routers 7-8, EIGRP on 9-10, and IS-IS on 11-12, as well as eBGP running between the autonomous systems.

IPV4:



IPV6:



Configurations

Router 7:

```
hostname Router7
ipv6 unicast-routing
interface Loopback0
 ip address 100.0.0.10 255.255.255.255
 ip ospf 1 area 0
 ipv6 address 2001:DB8:ACAD::10/128
 ipv6 ospf 1 area 0
interface GigabitEthernet0/0/0
 ip address 192.168.0.1 255.255.255.252
 ip ospf 1 area 0
 negotiation auto
 ipv6 address 2001:DB8:ACAD:1::1/64
 ipv6 ospf 1 area 0
 no shutdown
router ospfv3 1
 router-id 1.1.1.1
 address-family ipv6 unicast
 exit-address-family
router ospf 1
 router-id 1.1.1.1
```

```
banner motd ^CCCC UNAUTHORIZED ACCESS TO THIS DEVICE IS
PROHIBITED! This device has been configured for OSPF as a part
of the eBGP Cisco Lab ^C
End
```

Router 8:

```
hostname Router8
ipv6 unicast-routing
interface GigabitEthernet0/0/0
 ip address 192.168.1.1 255.255.255.252
 negotiation auto
 ipv6 address 2001:DB8:ACAD:2::1/64
 no shutdown
interface GigabitEthernet0/0/1
 ip address 192.168.0.2 255.255.255.252
 ip ospf 2 area 0
 negotiation auto
 ipv6 address 2001:DB8:ACAD:1::2/64
 ipv6 ospf 2 area 0
 no shutdown
router-id 2.2.2.2
address-family ipv6 unicast
 redistribute connected route-map BGP-TO-OSPF
 redistribute bgp 1 metric-type 1 route-map BGP-TO-OSPF
 router-id 2.2.2.2
exit-address-family
router ospf 2
 router-id 2.2.2.2
 redistribute connected subnets route-map BGP-TO-OSPF
 redistribute bgp 1 subnets route-map BGP-TO-OSPF
 network 192.168.0.0 0.0.0.3 area 0
router bgp 1
 bgp router-id 1.1.1.1
 bgp log-neighbor-changes
 neighbor 2001:DB8:ACAD:2::2 remote-as 2
 neighbor 192.168.1.2 remote-as 2
address-family ipv4
 network 192.168.1.0 mask 255.255.255.252
 redistribute ospf 2 route-map OSPF-TO-BGP
 no neighbor 2001:DB8:ACAD:2::2 activate
 neighbor 192.168.1.2 activate
 neighbor 192.168.1.2 next-hop-self
 neighbor 192.168.1.2 weight 100
 neighbor 192.168.1.2 route-map LP in
exit-address-family
```

```

address-family ipv6
  redistribute ospf 2 route-map OSPF-TO-BGP include-connected
  network 2001:DB8:ACAD:2::/64
  neighbor 2001:DB8:ACAD:2::2 activate
  neighbor 2001:DB8:ACAD:2::2 next-hop-self
  neighbor 2001:DB8:ACAD:2::2 route-map LP in
exit-address-family
route-map LP permit 10
  set local-preference 100
route-map BGP-TO-OSPF permit 10
  set metric 50
  set metric-type type-1
route-map OSPF-TO-BGP permit 10
banner motd ^CCCC UNAUTHORIZED ACCESS TO THIS DEVICE IS
PROHIBITED! This device has been configured for OSPF as a part
of the eBGP Cisco Lab ^C
end

```

Router 9:

```

hostname Router9
ipv6 unicast-routing
interface GigabitEthernet0/0/0
  ip address 192.168.2.1 255.255.255.252
  negotiation auto
  ipv6 address 2001:DB8:ACAD:3::1/64
  ipv6 eigrp 10
  no shutdown
interface GigabitEthernet0/0/1
  ip address 192.168.1.2 255.255.255.252
  negotiation auto
  ipv6 address 2001:DB8:ACAD:2::2/64
  no shutdown
router eigrp 10
  network 100.100.0.100 0.0.0.0
  network 192.168.2.0 0.0.0.3
  redistribute bgp 2 route-map BGP-TO-EIGRP
  redistribute connected route-map BGP-TO-EIGRP
router bgp 2
  bgp router-id 2.2.2.2
  bgp log-neighbor-changes
  neighbor 2001:DB8:ACAD:2::1 remote-as 1
  neighbor 192.168.1.1 remote-as 1
  address-family ipv4
    redistribute eigrp 10 route-map EIGRP-TO-BGP
    no neighbor 2001:DB8:ACAD:2::1 activate

```

```

neighbor 192.168.1.1 activate
neighbor 192.168.1.1 next-hop-self
neighbor 192.168.1.1 weight 50
neighbor 192.168.1.1 route-map LP in
exit-address-family
address-family ipv6
  redistribute eigrp 10 route-map EIGRP-TO-BGP include-connected
  neighbor 2001:DB8:ACAD:2::1 activate
  neighbor 2001:DB8:ACAD:2::1 next-hop-self
  neighbor 2001:DB8:ACAD:2::1 route-map LP in
exit-address-family
ipv6 router eigrp 10
  eigrp router-id 3.3.3.3
  redistribute bgp 2 route-map BGP-TO-EIGRP
  redistribute connected route-map BGP-TO-EIGRP
route-map BGP-TO-EIGRP permit 10
  set metric 1000000 10 255 1 1500
route-map EIGRP-TO-BGP permit 10
route-map LP permit 10
  set local-preference 50
banner motd ^CCCCCCCCC UNAUTHORIZED ACCESS TO THIS DEVICE IS
PROHIBITED! This device has been configured for EIGRP as a part
of the eBGP Cisco Lab ^C
end

```

Router 10:

```

hostname Router10
ipv6 unicast-routing
interface GigabitEthernet0/0/0
  ip address 192.168.3.1 255.255.255.252
  negotiation auto
  no shutdown
  ipv6 address 2001:DB8:ACAD:4::1/64
interface GigabitEthernet0/0/1
  ip address 192.168.2.2 255.255.255.252
  negotiation auto
  ipv6 address 2001:DB8:ACAD:3::2/64
  no shutdown
  ipv6 eigrp 10
router eigrp 10
  network 192.168.2.0 0.0.0.3
  redistribute bgp 3 route-map BGP-TO-EIGRP
  redistribute connected route-map BGP-TO-EIGRP
router bgp 3
  bgp router-id 3.3.3.3
  bgp log-neighbor-changes
  neighbor 2001:DB8:ACAD:4::2 remote-as 4

```

```

neighbor 192.168.3.2 remote-as 4
address-family ipv4
  network 192.168.3.0 mask 255.255.255.252
  redistribute connected
  redistribute eigrp 10 route-map EIGRP-TO-BGP
  no neighbor 2001:DB8:ACAD:4::2 activate
  neighbor 192.168.3.2 activate
  neighbor 192.168.3.2 next-hop-self
  neighbor 192.168.3.2 weight 100
  neighbor 192.168.3.2 route-map LP in
exit-address-family
address-family ipv6
  redistribute eigrp 10 route-map EIGRP-TO-BGP include-connected
  network 2001:DB8:ACAD:4::/64
  neighbor 2001:DB8:ACAD:4::2 activate
  neighbor 2001:DB8:ACAD:4::2 next-hop-self
  neighbor 2001:DB8:ACAD:4::2 route-map LP in
exit-address-family
ipv6 router eigrp 10
  eigrp router-id 4.4.4.4
  redistribute bgp 3 route-map BGP-TO-EIGRP
  redistribute connected route-map BGP-TO-EIGRP
route-map BGP-TO-EIGRP permit 10
  set metric 1000000 10 255 1 1500
route-map EIGRP-TO-BGP permit 10
route-map LP permit 10
  set local-preference 100
banner motd ^CCCC UNAUTHORIZED ACCESS TO THIS DEVICE IS
PROHIBITED! This device has been configured for OSPF as a part
of the eBGP Cisco Lab ^C
end

```

Router 11:

```

hostname Router11
ipv6 unicast-routing
interface GigabitEthernet0/0/0
  ip address 192.168.4.1 255.255.255.252
  ip router isis 1
  negotiation auto
  ipv6 address 2001:DB8:ACAD:5::1/64
  ipv6 router isis 1
  no shutdown
interface GigabitEthernet0/0/1
  ip address 192.168.3.2 255.255.255.252
  negotiation auto

```



```

ipv6 address 2001:DB8:ACAD:4::2/64
no shutdown
router isis 1
net 49.0001.0000.0000.0001.00
is-type level-1
metric-style narrow
log-adjacency-changes
redistribute connected route-map BGP-TO-IS-IS level-1
redistribute bgp 4 route-map BGP-TO-IS-IS level-1
address-family ipv6
    redistribute connected level-1 route-map BGP-TO-IS-IS
    redistribute bgp 4 level-1 route-map BGP-TO-IS-IS
exit-address-family
router bgp 4
bgp router-id 4.4.4.4
bgp log-neighbor-changes
neighbor 2001:DB8:ACAD:4::1 remote-as 3
neighbor 192.168.3.1 remote-as 3
address-family ipv4
    network 192.168.3.0 mask 255.255.255.252
    redistribute connected route-map IS-IS-TO-BGP
    redistribute isis 1 level-1 route-map IS-IS-TO-BGP
    no neighbor 2001:DB8:ACAD:4::1 activate
    neighbor 192.168.3.1 activate
    neighbor 192.168.3.1 next-hop-self
    neighbor 192.168.3.1 weight 50
    neighbor 192.168.3.1 route-map LP in
exit-address-family
address-family ipv6
    redistribute isis 1 level-1 route-map IS-IS-TO-BGP include-
connected
    network 2001:DB8:ACAD:4::/64
    neighbor 2001:DB8:ACAD:4::1 activate
    neighbor 2001:DB8:ACAD:4::1 next-hop-self
exit-address-family
route-map LP permit 10
    set local-preference 50
route-map IS-IS-TO-BGP permit 10
route-map BGP-TO-IS-IS permit 10
    set metric 50
    set metric-type external
banner motd ^CCCC UNAUTHORIZED ACCESS TO THIS DEVICE IS
PROHIBITED! This device has been configured for OSPF as a part
of the eBGP Cisco Lab ^C
end

```

Router 12:

```

hostname Router12
ipv6 unicast-routing
interface Loopback1
 ip address 100.0.0.20 255.255.255.255
 ipv6 address 2001:DB8:ACAD::20/128
interface GigabitEthernet0/0/0
 no ip address
 shutdown
 negotiation auto
interface GigabitEthernet0/0/1
 ip address 192.168.4.2 255.255.255.252
 ip router isis 1
 negotiation auto
 ipv6 address 2001:DB8:ACAD:5::2/64
 ipv6 router isis 1
 no shutdown
router isis 1
 net 49.0001.0000.0000.0002.00
 is-type level-1
 log-adjacency-changes
 passive-interface Loopback1
banner motd ^CCCC UNAUTHORIZED ACCESS TO THIS DEVICE IS
PROHIBITED! This device has been configured for OSPF as a part
of the eBGP Cisco Lab ^C
End

```

Proof:

```
<Router7#traceroute 2001:db8:acad:0::20>
```

Type escape sequence to abort.

```
Tracing the route to 2001:DB8:ACAD::20
```

```

 1 2001:DB8:ACAD:1::2 2 msec 1 msec 0 msec
 2 2001:DB8:ACAD:2::2 1 msec 1 msec 0 msec
 3 2001:DB8:ACAD:3::2 1 msec 1 msec 1 msec
 4 2001:DB8:ACAD:4::2 1 msec 1 msec 1 msec
 5 2001:DB8:ACAD:5::2 2 msec 2 msec 1 msec

```

```
Router7#ping 2001:db8:acad:0::20
```

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD::20, timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/3 ms

<Router7#traceroute 100.0.0.20>

Type escape sequence to abort.

Tracing the route to 100.0.0.20

VRF info: (vrf in name/id, vrf out name/id)

1 192.168.0.2 1 msec 1 msec 1 msec

2 192.168.1.2 1 msec 1 msec 0 msec

3 192.168.2.2 0 msec 1 msec 2 msec

4 192.168.3.2 1 msec 1 msec 1 msec

5 192.168.4.2 1 msec 1 msec *

Router7#ping 100.0.0.20

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 100.0.0.20, timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms

<Router12#traceroute 100.0.0.10>

Type escape sequence to abort.

Tracing the route to 100.0.0.10

VRF info: (vrf in name/id, vrf out name/id)

1 192.168.4.1 2 msec 1 msec 0 msec

2 192.168.3.1 1 msec 1 msec 1 msec

3 192.168.2.1 2 msec 1 msec 1 msec

4 192.168.1.1 1 msec 2 msec 1 msec

```
5 192.168.0.1 2 msec 2 msec *
Router12#ping 100.0.0.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.0.0.10, timeout is 2
seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max =
1/1/2 ms
<Router12#traceroute 2001:db8:acad:0::10>
Type escape sequence to abort.
Tracing the route to 2001:DB8:ACAD::10
```

```
1 2001:DB8:ACAD:5::1 1 msec 2 msec 1 msec
2 2001:DB8:ACAD:4::1 2 msec 2 msec 1 msec
3 2001:DB8:ACAD:3::1 1 msec 2 msec 1 msec
4 2001:DB8:ACAD:2::1 2 msec 2 msec 1 msec
5 2001:DB8:ACAD:1::1 1 msec 2 msec 2 msec
Router12#ping 2001:db8:acad:0::10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD::10, timeout is
2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max =
1/1/2 ms
```

Goodies:

```
<Router8#show ip bgp>
BGP table version is 16, local router ID is 1.1.1.1
Status codes: s suppressed, d damped, h history, * valid, >
best, i - internal,
                r RIB-failure, S Stale, m multipath, b backup-
path, f RT-Filter,
```

x best-external, a additional-path, c RIB-
 compressed,
 t secondary path, L long-lived-stale,
 Origin codes: i - IGP, e - EGP, ? - incomplete
 RPKI validation codes: V valid, I invalid, N Not found

Path	Network	Next Hop	Metric	LocPrf	Weight
*>	100.0.0.10/32	192.168.0.1	2		32768
?					
*>	100.0.0.20/32	192.168.1.2	5376	100	100
2 ?					
*>	192.168.0.0/30	0.0.0.0	0		32768
?					
*	192.168.1.0/30	192.168.1.2	0	100	100
2 ?					
*>		0.0.0.0	0		32768
i					
*>	192.168.2.0/30	192.168.1.2	0	100	100
2 ?					
*>	192.168.3.0/30	192.168.1.2	5376	100	100
2 ?					
*>	192.168.4.0/30	192.168.1.2	5376	100	100
2 ?					

<Router8#show bgp ipv6 unicast>
 BGP table version is 25, local router ID is 1.1.1.1
 Status codes: s suppressed, d damped, h history, * valid, >
 best, i - internal,
 r RIB-failure, S Stale, m multipath, b backup-
 path, f RT-Filter,
 x best-external, a additional-path, c RIB-
 compressed,
 t secondary path, L long-lived-stale,
 Origin codes: i - IGP, e - EGP, ? - incomplete
 RPKI validation codes: V valid, I invalid, N Not found

Path	Network	Next Hop	Metric	LocPrf	Weight
*>	2001:DB8:ACAD::10/128	FE80::521C:B0FF:FE2C:5100			
			1		32768
?					
*>	2001:DB8:ACAD::20/128	2001:DB8:ACAD:2::2			

```

2 ?
*> 2001:DB8:ACAD:1::/64
    :: 0 32768 ?
*> 2001:DB8:ACAD:2::/64
    :: 0 32768 i
*> 2001:DB8:ACAD:3::/64
    2001:DB8:ACAD:2::2
    0 100 0
2 ?
Network Next Hop Metric LocPrf Weight
Path
*> 2001:DB8:ACAD:4::/64
    2001:DB8:ACAD:2::2
    5376 100 0
2 ?
*> 2001:DB8:ACAD:5::/64
    2001:DB8:ACAD:2::2
    5376 100 0
2 ?

```

<Router7#show ip route>

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```

100.0.0.0/32 is subnetted, 2 subnets
C      100.0.0.10 is directly connected, Loopback0
O E1    100.0.0.20 [110/51] via 192.168.0.2, 00:17:04, GigabitEthernet0/0/0
192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks

```

```

C      192.168.0.0/30 is directly connected,
GigabitEthernet0/0/0
L      192.168.0.1/32 is directly connected,
GigabitEthernet0/0/0
      192.168.1.0/30 is subnetted, 1 subnets
O E1    192.168.1.0 [110/51] via 192.168.0.2, 00:17:31,
GigabitEthernet0/0/0
      192.168.2.0/30 is subnetted, 1 subnets
O E1    192.168.2.0 [110/51] via 192.168.0.2, 00:17:31,
GigabitEthernet0/0/0
      192.168.3.0/30 is subnetted, 1 subnets
O E1    192.168.3.0 [110/51] via 192.168.0.2, 00:17:31,
GigabitEthernet0/0/0
      192.168.4.0/30 is subnetted, 1 subnets
O E1    192.168.4.0 [110/51] via 192.168.0.2, 00:17:31,
GigabitEthernet0/0/0

```

```
<Router7#show ipv6 route>
```

```
IPv6 Routing Table - default - 9 entries
```

```
Codes: C - Connected, L - Local, S - Static, U - Per-user Static
route
```

```
      B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
```

```
      IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX -
EIGRP external
```

```
      ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr
- Redirect
```

```
      O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 -
OSPF ext 2
```

```
      ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, la - LISP
alt
```

```
      lr - LISP site-registrations, ld - LISP dyn-eid, lA -
LISP away
```

```
      le - LISP extranet-policy, a - Application
```

```
LC 2001:DB8:ACAD::10/128 [0/0]
```

```
      via Loopback0, receive
```

```
OE1 2001:DB8:ACAD::20/128 [110/51]
```

```
      via FE80::521C:B0FF:FE63:3831, GigabitEthernet0/0/0
```

```
C 2001:DB8:ACAD:1::/64 [0/0]
```

```
      via GigabitEthernet0/0/0, directly connected
```

```
L 2001:DB8:ACAD:1::1/128 [0/0]
```

```
      via GigabitEthernet0/0/0, receive
```

```
OE1 2001:DB8:ACAD:2::/64 [110/51]
```

```
      via FE80::521C:B0FF:FE63:3831, GigabitEthernet0/0/0
```

```
OE1 2001:DB8:ACAD:3::/64 [110/51]
```

```
      via FE80::521C:B0FF:FE63:3831, GigabitEthernet0/0/0
```

```
OE1 2001:DB8:ACAD:4::/64 [110/51]
```

```
    via FE80::521C:B0FF:FE63:3831, GigabitEthernet0/0/0
OE1 2001:DB8:ACAD:5::/64 [110/51]
    via FE80::521C:B0FF:FE63:3831, GigabitEthernet0/0/0
L   FF00::/8 [0/0]
    via Null0, receive
```

<Router8#show ip route>

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
    100.0.0.0/32 is subnetted, 2 subnets
O      100.0.0.10 [110/2] via 192.168.0.1, 00:22:28,
GigabitEthernet0/0/1
B      100.0.0.20 [20/5376] via 192.168.1.2, 00:22:02
    192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.0.0/30 is directly connected,
GigabitEthernet0/0/1
L      192.168.0.2/32 is directly connected,
GigabitEthernet0/0/1
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.1.0/30 is directly connected,
GigabitEthernet0/0/0
L      192.168.1.1/32 is directly connected,
GigabitEthernet0/0/0
    192.168.2.0/30 is subnetted, 1 subnets
B      192.168.2.0 [20/0] via 192.168.1.2, 00:30:53
    192.168.3.0/30 is subnetted, 1 subnets
B      192.168.3.0 [20/5376] via 192.168.1.2, 00:30:53
    192.168.4.0/30 is subnetted, 1 subnets
B      192.168.4.0 [20/5376] via 192.168.1.2, 00:22:32
```

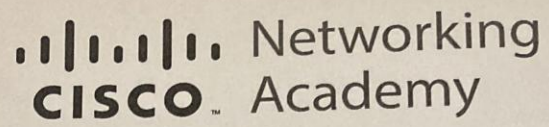

Problems

This lab was inundated with problems, not all of them actually solvable nor explicable, seeing as they sometimes just disappeared like fog in the morning sun when we reloaded some of the devices. One instance of that was when all the adjacencies failed despite the fact that all of the ports being used had “no shut” on them, but the second we reloaded the device the problem vanished. Beyond minor issues like that, BGP and redistribution were just a nightmare to work with in general. Our initial configurations just failed outright, no adjacencies forming whatsoever, and even when we tried using route-maps, and prefix-lists, they still failed. When we finally figured out what to do, that being properly configured route-maps and not whatever we had been doing previous, it had taken finding an obscure source consisting of a senior network engineer answering how to do this kind of thing on a random forum a couple years ago. This however led to a new problem, that being pings don’t go to the 192.168.2.0 network, as in we can ping through it, but not to it. After trying to see if the issue was with the route-maps through permitting all traffic, we figured out that the issue was that the routes were directly connected, meaning that they weren’t using the routing protocols, but that was solved using the “redistribute connected” command. The final real group of problems that we had was that only ipv4 pings worked throughout the entire network. We retried the redistribute connected for ipv6, but when that didn’t do anything we took a look at the whole configurations and figured out that Router 5 had been using level-2 IS-IS which was meant for internal routing, as opposed to what we should have been using, level-1 IS-IS which was for inter-area routing.

Conclusion

The lab was difficult to work through, especially due to a lack of specific resources that we could find online. Various protocol differences made the setup rather vexing, and the constant need to find new resources to determine if we were even using the right setups slowed the already grueling march. This was singularly most represented I believe in the realization that many of our problems on at least half of the connections could be solved with the simple application of the “redistribute connected” command, which we couldn’t at all find in any

sources when scouring the web, and instead had to resort to asking someone from another more advanced class entirely. What we did learn can be summarized in the following. Route-maps, as difficult and annoying to setup as they are, are highly important to the selection and filtering process of different autonomous systems when using BGP. Secondly, eBGP is a poorly documented, highly complicated mess of a protocol that nonetheless has managed to remain the favored protocol of the internet through reasoning beyond my ken.



Advanced Cisco Networking Academy Multi-Protocol Routing with eBGP

Kazu Li-Hirata

